

Sudipta Mandal

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Technical Skills & Methodologies

Backend: Laravel, Node.js, Express.js, Nest.js, REST, Django, GraphQL | **Frontend:** Next.js, React, MUI, HTML, CSS
DB: MySQL, Postgres, DynamoDB, Mongo, Eloquent, Prisma | **Cloud:** AWS (Lambda, EC2, Amplify, S3), Docker, GCP
Languages: JavaScript, TypeScript, PHP, C/C++, Java, Python | **Tools:** Git, Agile, Jira, Confluence, Jest, TDD, CI/CD
Architecture: System/API/DB Design, Caching, Event-driven Workflows, Background/Cron Jobs, AuthN/AuthZ, Redis
Agentic AI & Tooling: Context Engineering, MCP, Agentic Workflows, RAG, VectorDB, Claude Code, Cursor, Codex

Professional Working Experiences

Senior Software Engineer, [Cargo Stream](#) (SaaS [FreightTech](#), Vilnius, Lithuania, Remote) Nov 2025 - Present

- Architected and built a flexible, layout-aware document-processing system utilizing Laravel, pattern recognition, text analysis, and context mapping to accurately parse inconsistent logistic PDFs from multiple global partners.
- Designed custom extraction logic and robust fallback rules to seamlessly isolate tabular and irregular data from unstructured logistic documents, including orders, credit notes, and service invoices across multiple languages.
- Engineered automated normalization pipelines that transformed raw, varied PDF outputs into highly structured JSON and CSV formats, ensuring format-agnostic data integrity for downstream workflows with >99% accuracy.
- Leveraged agentic AI-assisted context engineering, pattern analysis, and agentic workflow automation to accelerate development workflows. Improved research and development engineering productivity by 50%.
- Collaborated closely with the CTO to optimize core data ingestion flows, significantly enhancing the parsing module's performance, robustness, accuracy, and scalability to support increasing document volumes.

Senior Software Engineer, [Sports Fixtures](#) ([Project-Based](#), Pattaya, Thailand, Remote) Oct 2025 - Jan 2026

- Built and maintained from scratch using Node.js, Strapi v5, PostgreSQL, and Redis. Deployed and managed production infrastructure with centralized logging, monitoring, and environment management across platforms.
- Designed and optimized 50+ relational database models and advanced multi-model filtering/query systems, improving large-scale data retrieval with caching and data indexing to increase performance by 40%.
- Implemented secure authentication and authorization features, including OTP verification, Google OAuth, Facebook OAuth, and JWT auth flow, improving login reliability and reducing authentication friction.
- Integrated complete [SportsDB](#) v1/v2 APIs, real-time live sports score functionality using WebSockets, and multiple third-party News APIs; optimized performance with Redis and Node cache strategies. Designed automated cron job schedulers and background workers that reduced API synchronization downtime by 30%.
- Boosted engineering productivity by 50% by architecting agentic AI workflows, automating documentation, and implementing advanced error logging and context-engineering strategies.

Software Engineer (L1, L2), [Eucaps AB](#) (SaaS [Fintech](#), Stockholm, Sweden, Remote) Dec 2021 - Oct 2025

- Built SaaS fintech features using Node.js, Express.js, Laravel, React/Next.js, Apollo/GraphQL, and REST APIs; leveraged AWS (Lambda, EC2, Amplify, S3), Docker, and Jest, with MySQL, PostgreSQL, and DynamoDB.
- Designed secure REST APIs (JWT + HMAC) enabling external partners to access selective database info; served 1K+ daily requests and cut integration time by 40%. Created clear and structured API documentation.
- Built backend pipelines to process 200MB+ [S&P](#) financial datasets (YoY, Margin, LTM) with cronjob schedulers, delivering low-latency GraphQL APIs that improved data accessibility for analysts and frontend functionalities.
- Integrated external APIs ([Pinpoint](#), [Inderes](#)) with cronjob schedulers, automating ingestion of 1K+ posts daily.
- Created scalable backend pipeline for 3K+ companies' financial charts & geo-maps with advanced filtering.
- Integrated [Stripe](#) payment and [Trulioo](#) KYC, enabling secure onboarding and seamless payment processing.
- Created backend pipelines to process data for 3K+ companies' events with services from [S&P](#) Capital IQ APIs.
- Leveraged GPT-3.5 + DALL·E to auto-generate hashtags and visuals for 3K+ company profiles, boosting analyst engagement by 25% and saving 10+ hrs/week in manual content preparation.

Junior Software Engineer, [NSUPS](#), (Voluntary, [Edtech](#), Dhaka, Bangladesh) Oct 2020 - Nov 2021

- Implemented multiple automatic crawlers to fetch data from 10+ online judges to track the solve count frequency of 100+ users. Created backend structure for managing the participants of NSUPS Bootcamp.
- Designed optimized database schemas and query structures to improve data consistency. Collaborated with team members to develop scalable backend features and support internal university programming events.

Agentic AI, Developer Tooling & Context Engineering

- Designed and optimized AI-assisted software development workflows using Claude Code, Antigravity, Cursor, Codex, and other agentic coding tools to accelerate feature development, debugging, and code maintenance.
- Utilized multiple frontier AI models, including Claude Opus, Sonnet, GPT, Composer, and Gemini, selecting model-specific workflows based on task complexity, reasoning requirements, and development objectives.
- Built custom context-engineering strategies leveraging persistent memory, structured context management, and MCP server integrations to automate repetitive development tasks and maximize AI agent reliability.
- Developed reusable AI-driven workflows for code generation, refactoring, documentation, testing, and project scaffolding, drastically minimizing manual engineering effort across professional and personal projects.

Personal Projects

Valo Dash ([SaaS Analytics Platform](#)), [System Design](#)

- Architected a multi-layer analytics dashboard for custom Valorant teams, using Node.js, TS, Next.js, OAuth, Zod, Prisma, and PostgreSQL. Integrated Riot Games APIs to collect and process player profiles and historical match data, powering team leaderboards, comparative analytics, graphical analysis, and performance insights.
- Implemented automated data refresh pipelines with scheduled cron jobs, maintaining up-to-date statistics and improving the reliability of backend data synchronization. Implemented a highly interactive data visualization UI.

Ask Sudipta ([Conversational Rag Intelligence](#)), [System Design](#)

- Built AskSudipta, a portfolio-integrated Conversational RAG assistant chatbot using React, TS, Express, Zod, Gemini, Supabase PostgreSQL, pgvector, and a vector database retrieval layer with noise reduction pipelines.
- Implemented markdown ingestion, heading-aware chunking, embeddings generation, hybrid semantic/vector search, keyword retrieval, grounded prompt construction, source citations, API auth, and response caching.

Color Cuddle ([Interactive Web Game](#)), [System Design](#)

- Developed Color Cuddle Games, an interactive web-based game using Next.js, featuring low-latency gameplay, drag-and-drop gameplay mechanics, responsive UI, and full mobile compatibility across devices.
- Optimized game logic, performance, accessibility, and user experience for both desktop and mobile platforms.

Text Analyzer ([Lexical Platform](#)), [System Design](#)

- Developed an online platform to analyze text data with auto-generated reports. Implemented CRUD for text, with each browser acting as a unique user. Ensured fast, lightweight analysis. Adopted by **10+** active users.
- Built an interactive UI, optimized analysis, and deployed using Node, Express, TS, React, Vite, MUI, and TDD.

DSE Ops ([Stock Analytics](#)), [System Design](#)

- Engineered a Node.js scraper to reliably extract daily SME company data from the Dhaka Stock Exchange.
- Parsed raw HTML tables into structured CSV files, automating data transformation for downstream analysis.

Research & Thesis

Social Media Opinion Mining Based on Bangla Public Post of Facebook ([Research](#)), [Detailed Overview](#)

- Research on sentiment analysis of Bangla text from Facebook public posts (Both Male & Female Celebrities)
- Coordinated data cleaning methods, data noise reduction, categorized, and annotated into 11 subcategories
- Created a dataset of 11K Bangla comments by using a **semi-automatic crawler** to fetch from Facebook
- Used state-of-the-art ML techniques (TF-IDF vectors, uni-bi, tri-gram, confusion matrix, etc) to analyze data
- Generated text clouds based on 3 main categories and 11 subcategories for the visual representation of data
- **Published** at the 24th ICCIT. **Skills learned:** Python, Google Colab, traditional ML algorithms, LaTeX, etc.

Education, Competitive Programming & Problem Solving

North South University (NSU), BSc in Computer Science & Engineering (CSE), **Graduated:** Fall, Aug 2021

- **Cumulative GPA: 3.53/4.0** (88%+), CSE Major CGPA: 3.62/4.00 (90%+), Graduate Distinction: **Cum Laude**
- ACM **ICPC** Dhaka Regional (2019-2020): Ranked [97th](#) among 190 teams (top 50%). Team: [NSU_0011](#)
- Top 100 at **NCPC**, MIST (2020-21). Intra NSU Programming Contest 2018: Ranked [10th](#) among 46 individuals
- Solved **650+ problems** in online judges. **Codeforces rating:** [Specialist, 1426](#) (Max) **Handle:** Mr_Mandal

Leadership & Communication Skills

- **Coordinator** of Junior Programming Training Program, **Mentored** 50+ students. **Problem Setter & Judge:** Intra-NSU Contest [Spring 2023](#), Bootcamp Graduation Contest [S07](#). PR officer in NSUPS'20 (Attendees: 1000+).
- Organized, judged, and set problems for multiple onsite & online contests organized by NSU and NSUPS. [Article](#)
- Invigilate 3+ **ICPC** preliminary contests at North South University. **English proficiency:** IELTS Speaking **Band 7**